Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given measurements and details. Begin your own position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A box at rest has **15 N** right and **3 N** left. **Find the additional horizontal force** needed for balance and **show the sum**.

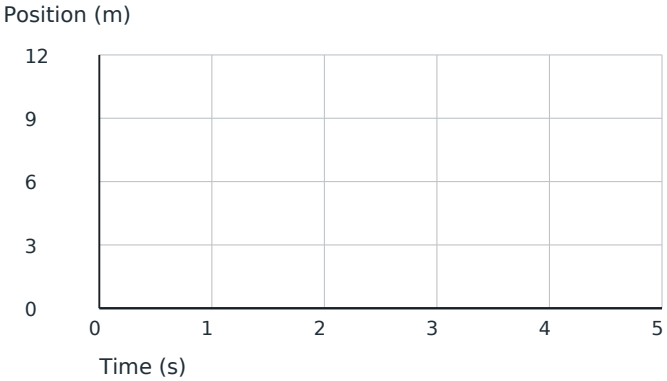
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Fictional fixed-Earth, constant-velocity calculation. CPA: x_i = **-120,000,000 km**; y_i = **0**; t = **16,000,000 s**; v_x = **8.21 km/s**; v_y = **2.20 km/s**.

Worked reasoning: A simplified calculation uses x_0 = **0 m** and constant v = **2 m/s** for **5 s**, giving x = **10 m**. This prediction depends on velocity staying constant. It does not prove that the real object moves that way.

Your turn: Draw the predicted path and **mark one observation you would** collect to **test the idea** that velocity stays constant.



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03 Organize the givens

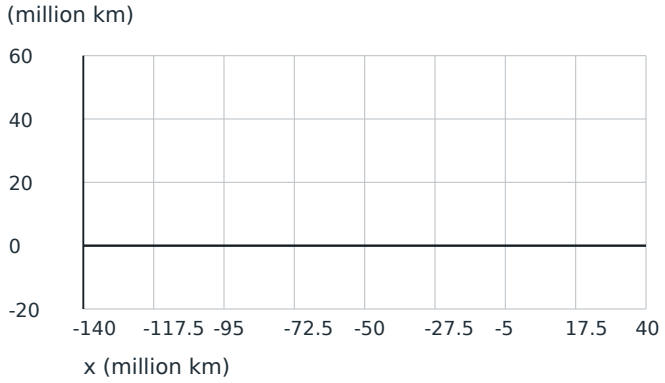
A fictional tracking exercise treats Earth as fixed at the origin. The object moves at constant velocity; gravity and Earth's motion are omitted.
1. Use the given values to **sketch its starting point** and both velocity directions.
2. Label +x toward Earth initially and +y perpendicular to it.

Quantity	Value
Initial x	-120,000,000 km
Initial y	0 km
Time	16,000,000 s
x velocity	+8.21 km/s
y velocity	+2.20 km/s

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04 Find the central end point

Using question **3**, find the change in each coordinate and then the final x and y coordinates. **Show each multiplication** with units. Plot the starting point and final point on the coordinate grid.



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Continue down each column.

05 Start the bounds

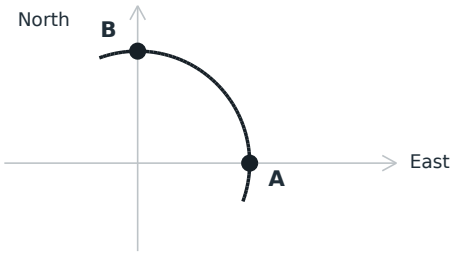
The same calculation uses independent ranges for its starting values: initial x from **-120,050,000** to **-119,950,000 km**; x velocity from **8.02** to **8.40 km/s**; y velocity from **2.15** to **2.25 km/s**. Initial y and time are exact. **Calculate lower** and upper final x and y coordinates.

Coordinate	Lower (km)	Upper (km)
x	___	___
y	___	___

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A rover travels straight *east* at a steady **7 m/s**. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.
Use the curve for the turning case; **show how velocity changes**.



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

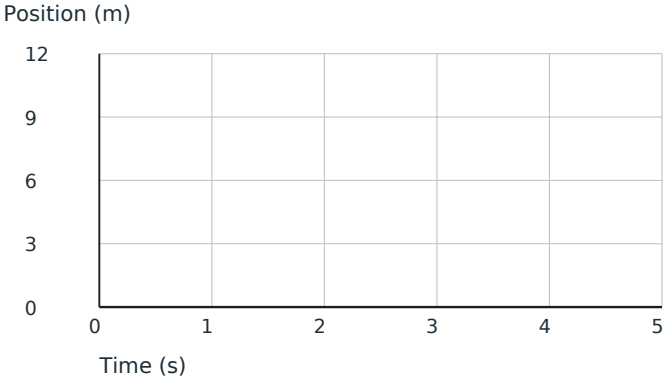
07 Checkpoint

State that your prediction treats velocity as constant. **Name one physical effect it** leaves out. **List what you finished** and your specific next step for Day **22**. Save all work.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Propose two different reasons a measured endpoint could be **8 m**.
2. **Give a measurement** that would help distinguish them.



Marzano check - circle: ① ② ③